The term 'robot' comes from a czech word, ‘robota’ meaning ‘forced labour’. Robotics is the science and study of robots.

What can Robots do?

Robots can sense and respond to their surroundings. They can handle delicate objects or apply great force. For example, they can perform eye operations guided by a human surgeon, or assemble a car. With **artificial intelligence**, robots will also be able to make decisions for themselves.

How do Robots sense?

Electronic sensors function as robot’s eyes and ears. Twin video cameras give the robot a 3-D view of the world. Microphones detect sounds. Pressure sensors give the robot a sense of touch, to judge how to grip an egg or heavy luggage. Built-in computers send and receive information with radio waves.



The quadrupedal military robot

Artificial Intelligence

Artificial intelligence attempts to create computer programs that think like human brains. Current research has not achieved this, but some computers can be programmed to recognize faces in a crowd.

Can Robots think?

Robots can think. They can play complex games, such as chess, better than human beings. But will a robot ever know that it is thinking? Humans are conscious - we know we are thinking. But we do not know how consciousness works. We do not know if Robots can ever be conscious.



Articulated welding robots (industrial)

Nanorobotics

Nanobots are robots scaled down to microscopic size in order to put them into very small spaces to perform a function. Future nanobots could be placed in the blood stream to perform surgical procedures that are too delicate or too difficult for standard surgery. Imagine if a nanobot could target cancer cells and destroy them without touching healthy cells nearby.



Future of Nanorobotics

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